Shared input streams

Coordinator parameters, layout, deterministic reduction

genomeGenerate workers

mutexInRead one worker parses and copies a whole chunk while locked

Prefix count and scatter

Adaptive record-safe chunks 1 MB at 64 or more threads

Independent suffix-bin sort

SAindex event scan

Junction search and insertion

Ordered deterministic merge / reduction

Bounded per-thread temporary memory

Recover allowed CPU-place union before pthread creation when required

Eligible private high-thread load: interleave allowed memory nodes Shared load: retain default policy

Shared read-mostly memory

Genome

Packed SA

SAindex

alignReads workers

Read chunk 1

Read chunk 2

Read chunk N